Confidence for Which Prediction? Supervision, Readout, and Coverage in Visual Grounding

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Abstract

Supervising a grounding confidence head to recognize correct outputs seems more direct than teaching it only whether a referent exists. Yet the same label change can have opposite reliability consequences. We study two frozen grounders with matched heads and training, preserving every output box. On FineCops validation, correct-output supervision changes mixed AUGRC by $+0.149$ *points for MM-GDINO and* -0.866 *for MDETR, despite improving correct-versus-wrong-box ranking in both. Reading the emitted query does not remove the MM-GDINO tradeoff; changing readout only at inference can differ from matched retraining, with substantial seed variation. Three-state analysis locates the MM-GDINO loss in comparisons involving complex positives outside the head's L1 supervision coverage. Within L1, both comparisons needed for successful-output ranking improve. Frozen transfer and fixed score combinations reveal when these findings extend or change. The results motivate inspecting which success/failure comparisons supervision changes, and how those comparisons relate to training coverage.*

1. Introduction

A visual grounder returning a box can fail in two ways: the requested referent is absent, or it exists but the chosen box is wrong. Generalized grounding makes both failures observable rather than guaranteeing a valid referent for every expression [3, 7, 8]. A confidence score used to retain reliable outputs must rank correct localizations above both failure types.

The natural training target is therefore *correct emission*: the referent exists and the emitted box is correct. If a score accurately estimated that success probability on the target population, ranking by it would favor successes at fixed coverage under equal failure costs. The empirical question concerns the finite heads we actually learn: *why can a more directly relevant supervision target still produce worse complete-output ranking?*

We isolate the label change by fixing the localizer, its output boxes, head architecture, initialization, requests and optimization. Existence and correct-emission heads differ only in the label assigned to present-but-wrong requests. The consequences differ across two frozen grounders. On FineCops validation, MM-GDINO's emission head improves correct-versus-wrong-box ordering but increases complete risk. MDETR's emission head improves that ordering and decreases complete risk. Both lose existence AUROC. An existence metric alone therefore obscures the comparison that determines whether reliable outputs were helped.

One grounding-specific explanation is a mismatch between the emitted box and the query supplying confidence. A Native score selects a box, but a global maximum over learned query logits may read another query. We test this explanation by training the same heads with a Native-selected-query readout. The MM-GDINO risk disadvantage survives. On MDETR, inference-only switching has a large adverse effect for one seed, whereas matched retraining gives smaller improvements in all three seeds. Reading the emitted query is consequently not a sufficient condition for resolving the observed reliability tradeoff.

The stronger explanatory lead is *which requests receive supervision*. The confidence loss uses 80,451 pairs drawn from 43,979 unique L1 positive parents, while validation also includes L2/L3 expressions. Within L1, MM-GDINO emission supervision improves both correct/wrong and correct/no-target comparisons. Its full-population correct/no-target loss lies in the cross-difficulty contribution involving L2/L3 positives and L1-parent negatives. Coverage and complex-expression generalization thus enter the main finding: the aggregate tradeoff does not characterize even all difficulty strata of this one localizer.

Our contribution is a controlled empirical account of a supervision change and the success/failure comparisons it alters. Matched and frozen-weight readout controls test a concrete explanation; three-state decomposition exposes a supervision-coverage boundary. Frozen transfer and sim-

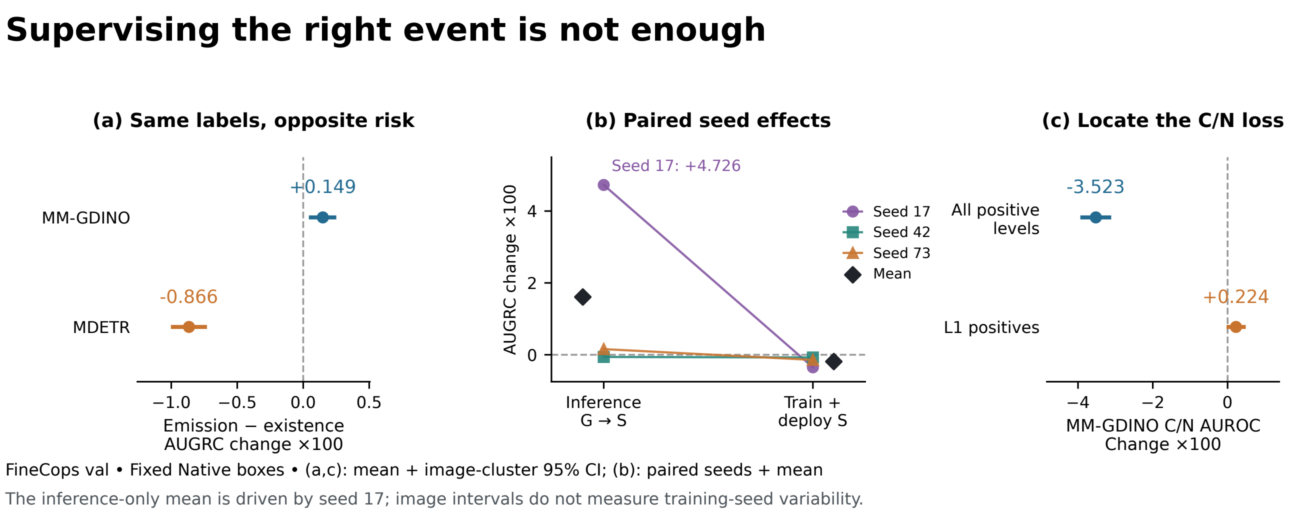

Figure 1. **From an endpoint difference to an explanation to investigate.** All panels use FineCops validation with fixed Native output boxes. (a) Changing existence labels to correct-emission labels raises complete risk on MM-GDINO and lowers it on MDETR. (b) For MDETR emission confidence, paired lines show all three seed effects relative to the same-seed G-trained/G-deployed head. The inference-only mean is driven by seed 17; matched retraining improves all three seeds. Diamonds denote means, not additional runs. (c) MM-GDINO's correct-output/no-target (C/N) target effect changes sign within L1, the only positive difficulty level supplying head-loss supervision. Panels (a,b) show AUGRC differences, where lower is better; (c) shows AUROC differences, where positive means a larger pairwise AUROC. Intervals in (a,c) resample images for the fixed three-head mean; they do not measure training-seed variability. Panel (b) displays seeds directly. The L1 comparison is not a training-coverage intervention.

ple score combinations then test practical consequences. A record-only evaluator accompanies the study so these comparisons can be applied to other fixed grounders. The experiment identifies the label effect under a fixed training source; testing coverage as its cause remains a distinct question.

2. Related Work

Grounding and prediction quality. Grounding DINO [9] and MDETR [5] provide language-conditioned queries and boxes. GREC/gRefCOCO add absent and multiple referents [3, 7]; FineCops evaluates compositional positives and negative expressions/images [8]. Localization-aware scores in detection [6] and learned reject options [1] connect confidence to prediction quality. We retain each grounder's Native box and study the confidence assigned to that fixed prediction.

Joint failure detection and score combination. Distinguishing invalid inputs and detecting wrong predictions are established as different problems [4]. OpenMix [14] and SCOD [10] study joint failure detection; SIRC analyzes confidence combination and changing failure mixtures [13]. AUGRC provides a generalized-risk view of selective prediction [12]. These works supply the broader framework for our three states and population-dependent score preferences. Our independent evidence comes from a matched label intervention, query-readout controls, and difficulty/same-image structure within visual grounding. The SIRC-style score is a fixed control using training statistics, rather than a newly proposed estimator.

Generalization and operating points. Calibration and its stability under distribution shift concern the interpretation of score values and thresholds [2, 11]. Here the primary outcome is ordering fixed outputs by success. Frozen-head transfer tests a learned ordering on another request population; no evaluation threshold is fitted. Training support is also explicit, so performance on complex expressions can be distinguished from the pair population supplying head supervision.

3. Matched Study

3.1. Fixed predictions and two targets

For image–expression request (I, e), a frozen localizer supplies query features q_j, Native scores s_j^0 and boxes b_j. The Native selector chooses j^* and emits $b^* = b_{j^*}$, preserving its original tie rule. We distinguish C (referent present, Native IoU ≥ 0.5), W (present but wrong), and N (no target). Existence target E is one on C, W; emission target Y is one only on C. No-target has no fabricated GT.

Every head computes dense logits $z_j = h([q_j; s_j^0])$ from

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detached inputs. The two readouts are

$$c_G = \max_{j \in \mathcal{V}} z_j, \qquad c_S = z_{j^*}. \tag{1}$$

Both training and matched inference use the declared readout. We also retain the alternate reading of each fixed head. Thus inference-only switching and matched retraining are separately observed. Retraining S changes query competition and gradient locations along with the readout; it starts from the original initialization, not the trained G endpoint. Confidence never selects the output box.

3.2. Localizers, training and coverage

The first localizer is a five-epoch positive-source FineCops MM-GDINO-T artifact with 900 256-d queries. The second is the official RefCOCO MDETR-R101 EMA model with 100 256-d queries and Native score $1 - p_{\text{no-object}}$. Each model supplies its own correctness labels. Their architectures, adaptation and Native accuracies differ; comparisons of target/readout effects are made *within* each localizer.

All heads have LayerNorm(257), GELU MLP $257 \to 128 \to 128 \to 1$, 50,179 parameters/eight tensors and zero-initialized final output. Seeds 17/42/73 share initialization and request order across matched arms. Five epochs give 12,575 updates, batch 32 pairs, deterministic FP32, AdamW LR 10^{-4}, WD zero and clip 0.1. The two sources each contribute half the mean BCE, plus source-averaged logit L2 with weight 10^{-3}. Emission labels receive no additional reweighting. Cached features are FP16; Native scores/boxes and head computation are FP32. Six MM-GDINO global heads are reused, with exact record parity; six selected MM-GDINO and twelve MDETR heads complete the matrix without localizer updates.

The 80,451 confidence pairs use 43,979 unique L1 positive parents; no L2/L3 positive enters the head loss. The complete 83,341-positive training cache supplies later combination statistics, not additional supervision. MM-GDINO's earlier trunk adaptation did use the full positive source. In validation, all 9,029 text negatives also have L1 parents, while positives span all three levels. These coverage facts are shared by every target/readout arm.

3.3. Populations, endpoints and uncertainty

FineCops validation has 9,426 positives and 9,029 text negatives. Frozen transfer uses gRefCOCO TestA/B: 11,563 single-target positives, 9,121 no-target requests and 1,500 images. All 14,579 multi-target expressions are excluded. Removing FineCops train/val source images through VG–COCO identities or exact bytes leaves 1,277 images, 9,848 positives and 7,716 no-targets. This sensitivity is not disjoint from all pretraining or MDETR's RefCOCO lineage. Full and source-disjoint surfaces overlap. FineCops Test and negative images are not evaluated anew.

Mixed AUGRC, integrating $\Pr(Y = 0, c \geq t)$ over coverage, is the primary complete-output metric. We retain AURC, pairwise AUROCs, Native P@1, fixed-coverage error composition and diagnostic FPR95. Risk and AUROC differences below are multiplied by 100. The 5,000 paired image-cluster draws use PCG64 seed 20260911, shared across all scores, localizers and head seeds on a surface; gRef draws are stratified by TestA/B. Metrics are recomputed per seed, then equally averaged. Intervals describe the fixed three-head mean under image resampling; sample SD separately describes head-seed variation. Every FPR95 replicate recomputes positive q05. This is a post-hoc-motivated study with a prospectively fixed recipe, without checkpoint selection or deployment-threshold fitting.

4. Same Supervision, Opposite Risk Results

The global emission-minus-existence change on FineCops is +0.149 [+0.058, +0.235] AUGRC points for MM-GDINO, but -0.866 [-0.988, -0.745] for MDETR. Both improve correct-versus-wrong-box AUROC, by 11.654 and 13.889 points respectively, and both lose existence AUROC. Matched training and fixed outputs make this a directly observed supervision effect, yet the complete-risk direction is localizer-dependent (Fig. 1a). Native P@1 is 77.36% and 65.56%, respectively, and remains unchanged for every confidence route.

5. Does Reading the Output Query Resolve It?

Selected readout changes MM-GDINO emission risk by -0.023 [-0.048, +0.003]; the emission-minus-existence disadvantage remains +0.159 [+0.077, +0.241]. Thus the FineCops tradeoff persists even when confidence reads the actual emitted query. On MDETR, selected readout changes emission risk by -0.189 [-0.257, -0.122], while existence improves by a similar amount. Table 1 retains both absolute effects $D_Y = R_{S,Y} - R_{G,Y}$, $D_E = R_{S,E} - R_{G,E}$, and $I = D_Y - D_E$. FineCops interactions are +0.011 [-0.016, +0.038] and +0.001 [-0.091, +0.088]: neither resolves an emission-specific readout advantage.

Re-reading is different from retraining. MDETR's globally trained emission head incurs +1.608 [+1.513, +1.698] AUGRC points when only its inference reading changes to Selected. The seed effects are $+4.726$, -0.061, and $+0.158$ for seeds 17/42/73 (sample SD 2.703): seed 17 drives the large mean. Matched training/deployment of Selected instead changes risk by -0.189 [-0.257, -0.122] relative to the same G/G reference (Table 2), with seed effects -0.349, -0.075, and -0.143. Figure 1(b) shows these paired effects; the narrow image interval for the mean is not evidence of low training-seed variability. These interventions distinguish an inference change from a retrained

Table 1. Target by readout with unchanged Native boxes. Mixed AUGRC $\times 100$ (lower is better). The four cells show seed mean (sample SD); effects show mean [paired 95% CI]. $D_Y = R_{S,Y} - R_{G,Y}$, $D_E = R_{S,E} - R_{G,E}$, and $I = D_Y - D_E$. The gRef surfaces overlap and are not independent replications.

Localizer	Population	G/E	G/Y	S/E	S/Y	D_Y	D_E	I
MM-GDINO-T	FineCops val	26.23 (0.07)	26.38 (0.04)	26.20 (0.07)	26.36 (0.04)	-0.023 [-0.048, +0.003]	-0.034 [-0.068, +0.000]	+0.011 [-0.016, +0.038]
MM-GDINO-T	gRef source-disjoint	28.68 (0.08)	28.61 (0.04)	28.65 (0.11)	28.63 (0.05)	+0.022 [-0.013, +0.057]	-0.027 [-0.060, +0.007]	+0.049 [+0.013, +0.085]
MM-GDINO-T	gRef Full	28.79 (0.08)	28.75 (0.05)	28.76 (0.11)	28.76 (0.05)	+0.011 [-0.021, +0.045]	-0.025 [-0.055, +0.006]	+0.037 [+0.003, +0.069]
MDETR-R101	FineCops val	29.71 (0.27)	28.84 (0.13)	29.52 (0.28)	28.65 (0.02)	-0.189 [-0.257, -0.122]	-0.191 [-0.285, -0.094]	+0.001 [-0.091, +0.088]
MDETR-R101	gRef source-disjoint	20.11 (0.54)	19.92 (0.38)	20.34 (0.15)	20.35 (0.06)	+0.437 [+0.363, +0.507]	+0.236 [+0.137, +0.332]	+0.201 [+0.108, +0.292]
MDETR-R101	gRef Full	20.25 (0.53)	20.09 (0.39)	20.52 (0.14)	20.53 (0.06)	+0.440 [+0.373, +0.506]	+0.268 [+0.176, +0.356]	+0.172 [+0.085, +0.259]

Table 2. Two ways to read the Native-selected query on MDETR FineCops. Both changes reference the G-trained/G-deployed emission head. Matched S is trained from the same original initialization, not continued from G. Risk is mixed AUGRC times 100; endpoint cells show seed mean (sample SD), differences show paired image intervals for the fixed three-head mean.

Training	Inference	Risk (SD)	Change from G/G
Global	Global	28.84 (0.13)	–
Global	Selected	30.45 (2.83)	+1.608 [+1.513, +1.698]
Selected	Selected	28.65 (0.02)	-0.189 [-0.257, -0.122]

system along a specified path, rather than uniquely decomposing spatial and optimization contributions.

Winner geometry confirms that the two decisions sometimes use different evidence. MM-GDINO's global-emission winner agrees with Native's query on about 95% of C and 82% of W; about 9% of Native-wrong positives have a spatially correct confidence winner. The supplement reports box overlap and all alternate readouts. For an S-trained head, the global winner is a counterfactual diagnostic; deployed S always reads Native.

6. Which Comparisons Determine Complete Risk?

Let U_{CW}, U_{CN}, U_{WN} denote pairwise AUROCs with half credit for ties, and $a = \Pr(C \mid E = 1)$. Existence AUROC decomposes as

$$\mathrm{AUC}_E = aU_{CN} + (1-a)U_{WN}. \tag{2}$$

The W/N term orders two failures. Complete success ranking instead depends on C/W and C/N. At no-target prior π, fixed predictions give the identity

$$\Delta R = -(1-\pi)a\big[(1-\pi)(1-a)\Delta U_{CW} + \pi\Delta U_{CN}\big]. \tag{3}$$

Here R is AUGRC, not AURC. Prior reweighting holds class-conditional requests fixed. Interior crossover roots and their absence are reported in the supplement.

On MM-GDINO FineCops, emission supervision improves C/W by 11.654 points but changes C/N by -3.523 [-3.884, -3.166]. Their observed-mixture risk contributions are -0.532 and $+0.681$ points, summing to $+0.149$. On MDETR, C/W improves by 13.889 points while the C/N change, $+0.290$ $[-0.265, +0.857]$, is small and unresolved. The corresponding contributions are -0.818 and -0.047, summing to -0.866. MDETR's large W/N loss reduces existence AUROC without directly increasing complete risk. These comparisons explain why the same existence-metric direction has different endpoint consequences.

The distinction also appears among accepted requests. At approximately 50% coverage on MM-GDINO FineCops, changing from global existence to global emission reduces the accepted wrong-box fraction from 8.44% to 3.55%, but raises the accepted no-target fraction from 41.46% to 46.81%. Both denominators are accepted requests. The two failure components, rather than a single validity AUROC, identify the cost.

Table 3. The same existence-AUROC change need not imply the same output-risk change. Emission-minus-existence effects $\times 100$ [paired 95% CI]; positive denotes larger AUROC, negative denotes lower AUGRC. C/W tests localized correctness, C/N tests successful outputs against absent targets, and W/N compares two failures. The first two determine fixed-population AUGRC, whereas existence AUROC also includes the third. Readout contrasts and gRef Full are retained in Table 1 and the complete analysis.

Localizer	Population	Readout	C/W	C/N	W/N	Existence	AUGRC
MM-GDINO-T	FineCops val	G	+11.654 [+10.721, +12.641]	-3.523 [-3.884, -3.166]	-17.795 [-18.821, -16.781]	-6.754 [-7.211, -6.308]	+0.149 [+0.058, +0.235]
MM-GDINO-T	FineCops val	S	+10.724 [+9.850, +11.606]	-3.359 [-3.700, -3.019]	-16.823 [-17.738, -15.898]	-6.408 [-6.827, -5.984]	+0.159 [+0.077, +0.241]
MM-GDINO-T	gRef source-disjoint	G	+6.533 [+5.605, +7.467]	-3.184 [-3.635, -2.751]	-7.204 [-7.895, -6.501]	-4.951 [-5.398, -4.511]	-0.066 [-0.178, +0.043]
MM-GDINO-T	gRef source-disjoint	S	+6.056 [+5.198, +6.931]	-3.270 [-3.706, -2.853]	-7.774 [-8.476, -7.074]	-5.250 [-5.706, -4.806]	-0.018 [-0.127, +0.090]
MDETR-R101	FineCops val	G	+13.889 [+13.044, +14.743]	+0.290 [-0.265, +0.857]	-14.257 [-15.124, -13.406]	-4.719 [-5.271, -4.165]	-0.866 [-0.988, -0.745]
MDETR-R101	FineCops val	S	+14.622 [+13.664, +15.565]	+0.017 [-0.577, +0.618]	-15.673 [-16.619, -14.698]	-5.386 [-5.977, -4.774]	-0.864 [-0.993, -0.737]
MDETR-R101	gRef source-disjoint	G	+6.204 [+5.153, +7.325]	-0.493 [-0.917, -0.082]	-7.990 [-9.622, -6.478]	-1.844 [-2.322, -1.395]	-0.189 [-0.291, -0.083]
MDETR-R101	gRef source-disjoint	S	+4.992 [+3.774, +6.338]	-1.211 [-1.707, -0.698]	-7.151 [-8.950, -5.463]	-2.281 [-2.833, -1.738]	+0.013 [-0.119, +0.138]

Table 4. The MM-GDINO FineCops global target effect changes across difficulty comparisons. All numbers are emission-minus-existence AUROC points with existing paired image intervals. L1 uses 6,097 positive requests and the same 9,029 no-target requests. The final two rows are weighted contributions to the full C/N effect, not conditional AUROCs.

Comparison	ΔU_{CW}	ΔU_{CN}
Full positive population	+11.654 [+10.721, +12.641]	-3.523 [-3.884, -3.166]
L1 positive population	+7.085 [+6.023, +8.236]	+0.224 [+0.024, +0.431]
Within-level contribution	–	+0.169 [+0.019, +0.326]
Cross-level contribution	–	-3.692 [-4.019, -3.384]

7. Supervision Coverage and Generalization

The full-population loss is not the L1 result. Table 4 identifies a sharper boundary than a generic population-dependence statement. Within L1, MM-GDINO's global target change improves C/W by +7.085 [+6.023, +8.236] and C/N by +0.224 [+0.024, +0.431]. The full-population C/N decrease comprises a within-level contribution of +0.169 [+0.019, +0.326] and a cross-level contribution of -3.692 [-4.019, -3.384]. Since every negative parent is L1, the latter compares L2/L3 correct positives against L1-parent no-target requests. These complex positives are outside the positive difficulty support of the confidence loss.

An analytic consequence of the existing means. Restrict the positive population to L1, retaining the existing no-target requests. Then $C = 5{,}506$, $W = 591$, and $a_1 = 5{,}506/6{,}097$. Both point differences, $\Delta U_{CW}^{(1)} = 0.070846$ and $\Delta U_{CN}^{(1)} = 0.002238$, are positive. Equation 3 therefore gives $\Delta R_1(\pi) < 0$ for every $0 \leq \pi < 1$, with equality at the all-no-target endpoint. The fixed equal-seed mean curve has no internal prior reversal on this conditional population. This is a post-hoc analytic implication of the reported pairwise means: no L1 risk bootstrap, new curve interval or simultaneous statistical guarantee has been computed.

The negative comparison also persists within images. Its C/N change is -1.163 [-1.557, -0.760], versus -1.700 [-2.002, -1.400] without conditioning on the same participating requests. Thus it is not solely a cross-image ordering effect. Difficulty conditioning changes the available state pairs; it is not an image-matched or exactly distribution-matched intervention. Together, these observations prioritize supervision coverage and complex-expression generalization as the next explanation to test. They do not identify coverage as the cause: no arm changed which difficulty levels received supervision. Positive difficulty and negative edit level are kept distinct, and gRef uses same-image comparisons rather than fabricated edited-expression pairs.

Table 5. Fixed combinations on the same Native boxes. Mixed AUGRC $\times 100$, and paired differences against all three prespecified references. Lower is better. Product and SIRC-style use Native and global-existence scores; only SIRC-style uses training-positive score statistics. No evaluation mixture, weight or threshold is fitted.

Localizer	Population	Score	AUGRC (SD)	Δ Native	$\Delta G/E$	$\Delta G/Y$
MM-GDINO-T	FineCops val	Product	26.62 (0.03)	-1.497 [-1.598, -1.398]	+0.386 [+0.286, +0.488]	+0.238 [+0.177, +0.302]
MM-GDINO-T	FineCops val	SIRC-style	27.35 (0.02)	-0.758 [-0.821, -0.696]	+1.125 [+0.972, +1.280]	+0.977 [+0.865, +1.090]
MM-GDINO-T	gRef source-disjoint	Product	28.48 (0.03)	-1.569 [-1.727, -1.407]	-0.196 [-0.282, -0.111]	-0.130 [-0.187, -0.074]
MM-GDINO-T	gRef source-disjoint	SIRC-style	28.71 (0.01)	-1.338 [-1.482, -1.191]	+0.035 [-0.088, +0.157]	+0.101 [+0.040, +0.164]
MDETR-R101	FineCops val	Product	29.33 (0.18)	-0.831 [-0.999, -0.663]	-0.381 [-0.439, -0.321]	+0.485 [+0.385, +0.589]
MDETR-R101	FineCops val	SIRC-style	30.01 (0.00)	-0.146 [-0.159, -0.134]	+0.304 [+0.107, +0.493]	+1.170 [+1.038, +1.302]
MDETR-R101	gRef source-disjoint	Product	19.84 (0.46)	-1.291 [-1.485, -1.088]	-0.265 [-0.295, -0.237]	-0.076 [-0.170, +0.015]
MDETR-R101	gRef source-disjoint	SIRC-style	20.95 (0.03)	-0.184 [-0.204, -0.156]	+0.843 [+0.644, +1.039]	+1.031 [+0.881, +1.179]

Frozen transfer tests the learned design. MDETR's Selected emission head, which improved FineCops risk, worsens gRef source-disjoint risk by +0.437 [+0.363, +0.507]; the loss is positive in all three head seeds. Full TestA/B gives the same direction. Its mean target effect and interaction are more seed-sensitive, with opposite signs across heads. Their image intervals concern the fixed three-head average, not training a fresh head. For MM-GDINO source-disjoint, emission's AURC advantage is resolved but its primary AUGRC comparison is not. All seeds, both gRef surfaces and both integrals remain reported.

8. Can Simple Combinations Alleviate the Tradeoff?

We close with two fixed combinations of Native score s_0 and the same-seed global-existence logit z_E:

$$c_{\text{product}} = \log \max(s_0, 10^{-6}) + \log \sigma(z_E), \tag{4}$$

$$c_{\text{SIRC}} = -\log \max(1 - s_0, 10^{-6}) - \operatorname{softplus}(-(z_E - (\mu - 3\sigma_E))/\sigma_E). \tag{5}$$

The SIRC-style control [13] uses mean and population SD on all 83,341 unique training positives; at $\sigma_E \leq 10^{-12}$ it falls back to Native. No evaluation-set weight or threshold is fitted.

On MM-GDINO gRef source-disjoint, Product improves over Native, global existence and global emission; the last difference is -0.130 [-0.187, -0.074] AUGRC points. On FineCops, however, it loses to both learned MM-GDINO heads despite improving over Native. For MDETR it improves over Native and existence, but loses to emission on FineCops; the emission comparison is unresolved on source-disjoint gRef, though favorable on Full. Table 5 shows these costs beside the gains. These fixed combinations demonstrate that learned information can be combined profitably in some settings; they neither define a new method contribution nor dominate every reference.

9. Discussion and Conclusion

The study replaces a generic supervision conflict with a more specific question: *which successful-output comparisons changed, on which part of the request population?* Reading the emitted query is not a sufficient repair, and inference re-reading can behave differently from matched training. Three-state analysis reveals that MM-GDINO's negative effect is concentrated in comparisons involving complex positives beyond the head's L1 supervision support, even though the within-L1 success comparisons improve. A coverage-controlled intervention is the natural next test of that explanation, rather than a conclusion of these data.

The evidence is bounded by two fixed localizers, three head seeds, negative-text/single-no-target requests and previously investigated benchmarks. Localizer lineage, complete GREC, negative images and calibrated operating points remain separate scopes. Within these bounds, the controls, decomposition, transfer and combination results connect an endpoint difference to concrete evidence useful for confidence design.

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